Rocks and Minerals

Brown Science Olympiad Division B
Invitational December 7, 2025

Instructions and Advice:

- **DO NOT WRITE ON THE EXAM**
- Allowed materials: one 3 ring binder of any size, one (1) magnifying glass, one (1) commercially produced field guide, one (1) copy of the 2026 National Rocks and Minerals List, and one (1) stand-alone, non-programmable, non-graphing calculator
- You do not have to show your work unless you are explicitly asked to do so. Partial credit may be awarded if correct work is shown in these cases.
- Write all answers on the answer sheets provided; answers written on the exam/image sheets will not be graded. Good luck!

Station 1

1. (1 point) Identify the following rock.



2. (3 points) What type of rock is it? Be as specific as possible naming potential subtypes.

3. (2 points) What two metal elements are these rocks usually associated with?

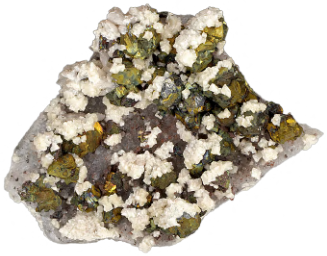
4. (1 point) Where in the United States is the rock mostly mined for?

5. (3 points) Explain the concept of laterization and how it relates to this rock. **(Tiebreaker #1)**

6. (2 points) Outline a possible pathway aligning with the rock cycle that might lead to the formation of this rock. Start from the initial formation of igneous rocks from the solidification of molten rock and go from there. Words or diagrams are fine.

Station 2

1. (1 point) Identify the mineral.



2. (2 points) What two other minerals are flanking the mineral?

3. (1 point) What class of mineral is it?

4. (1 point) What is the chemical formula of this mineral?

5. (1 point) What is the approximate hardness of this mineral?

6. (2 points) What would you expect to see if you performed a hardness test on the mineral using a knife? What about a copper coin?

7. (1 point) What element is this mineral a common ore of?

8. (1 point) Identify the mineral.



9. (1 point) What family does this mineral belong to?

10. (2 points) What environments are favorable for the formation of this mineral? Based on your answer, what types of rocks will typically show this mineral?
(Tiebreaker #2)

11. (1 point) Both of these minerals exhibit what kind of symmetry that allows for twinning?

Station 3

1. (1 point) Identify the following mineral.



2. (1 point) What distinct crystal habit does the image display?

3. (1 point) What is the streak of this mineral?

4. (1 point) Is this mineral magnetic? (Yes/No)

5. (1 point) What is the hardness of this mineral?

6. (1 point) Where in the United States is this mineral most commonly found?

7. (1 point) What common sedimentary rock/feature commonly displays this mineral?

8. (3 points) How does this mineral pertain to the exploration of Mars? What does the presence of this mineral on Mars indicate about the planet?

Station 4

1. (1 point) Identify the following.



2. (2 points) What type of fracture and cleavage does this mineral exhibit?
3. (1 point) What family does this mineral belong to?
4. (2 points) What two other minerals on the rocks minerals list are the most similar to this mineral?
5. (1 point) What element is this mineral particularly rich in?
6. (1 point) Which igneous rock is this mineral often found in?
7. (1 point) Based on your previous answer or your knowledge of the mineral, is this mineral likely felsic or mafic in origin?

8. (1 point) Identify the following mineral.



9. (1 point) What is the chemical formula of this mineral?
10. (1 point) What family does this mineral belong to?
11. (2 points) What is the luster of this sample of the mineral? What other luster does this mineral commonly display?
12. (2 points) Based on your answer for the previous question, give a chemical explanation for why these two lusters exist. Determine what other element gives the mineral its particular appearance.
13. (1 point) What element is this mineral a common ore of?
14. (1 point) What do you expect to see if you shine the mineral under UV light?

Station 5

1. (1 point) Identify the mineral.



2. (1 point) What is the chemical formula of this mineral?

3. (2 points) What type of rock are you most likely to find this mineral? What about the environment is favorable to the formation of this mineral?

4. (1 point) This mineral displays a unique feature of being able to pierce other minerals and rocks. What do you call this structure when it forms in quartz?

5. (1 point) What is the name of the crystal habit in the previous question?

6. (1 point) This mineral frequently displays twinning. What kind of symmetry allows for the twinning in this mineral?

7. (3 points) What are the 6 different symmetries/systems that can occur in different minerals that can allow twinning?
   (Tiebreaker #3)

8. (2 points) Describe the difference between contact and penetration twinning.

Station 6

1. (1 point) Identify the following.



2. (1 point) What is the chemical composition of the mineral?
3. (2 points) Explain the concept of polymorphism and connect it to this mineral. **(Tiebreaker #4)**
4. (1 point) Does this mineral exhibit perfect cleavage (Yes/No)?
5. (2 points) What element gives this mineral its green/blue color? What can you use to confirm the presence of this element?

6. (1 point) Identify the following



7. (1 point) What broad classification fits both this mineral and the previous mineral?

8. (1 point) This mineral primarily has more of which element compared to that of the previous mineral?

9. (3 points) What environments are favorable for the formation of this mineral? Based on your answer, what types of rocks will typically show this mineral?
(Tiebreaker #5)

10. (1 point) Both of these minerals exhibit what kind of symmetry that allows for twinning?

Station 7

1. (1 point) Identify the following.



2. (1 point) This mineral forms in what types of rocks?
3. (2 points) What is the streak and hardness of this mineral?
4. (1 point) What temperature does this mineral crystallize at in celsius?
5. (2 points) Fill in the blank. This mineral reacts with _____ to create _____.
6. (2 points) Describe the concept of fractional crystallization. Which elements are removed early on and which elements remain after a long period of time in this process?
7. (5 points) What's the difference between the continuous and discontinuous reaction series? Be as specific as possible and describe when the continuous and discontinuous reaction series converge.
   (Tiebreaker #6)
8. (1 point) Name two subtypes of this mineral and describe briefly how they differ chemically.

Station 8

1. (2 points) What two minerals are present in the sample?



2. (2 points) What is the chemical composition of both minerals?
3. (1 point) Will this mineral react with acid at low temperatures? What about high temperatures?
4. (2 points) How are these two minerals related when they occur together and how can one mineral lead to the formation of the other?
5. (1 point) These minerals are good indicators of what type of deposit?
6. (1 point) What is the hardness of both minerals?
7. (1 point) What crystal habit does the specimen show?

Station 9

1. (1 point) Identify the following.



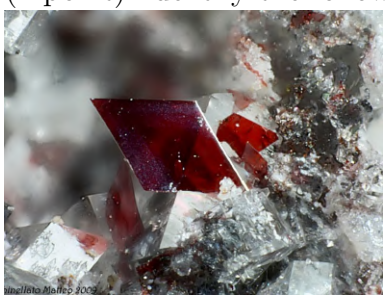
2. (1 point) Identify the following.



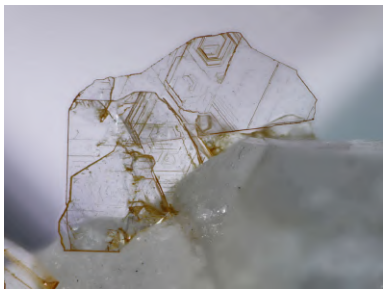
3. (1 point) Identify the following.



4. (1 point) Identify the following.



5. (1 point) Identify the following.



6. (1 point) Identify the following.



7. (1 point) Identify the following.

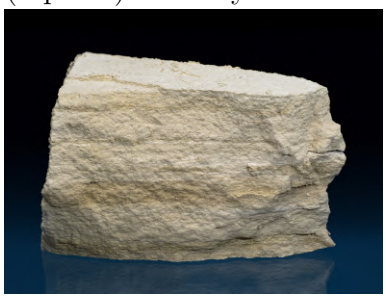


8. (1 point) Identify the following.



Station 10

1. (1 point) Identify the following.



2. (1 point) Identify the following.



3. (1 point) Identify the following.



4. (1 point) Identify the following.



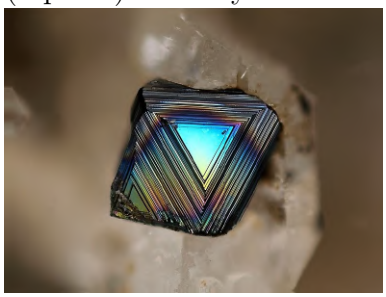
5. (1 point) Identify the following.



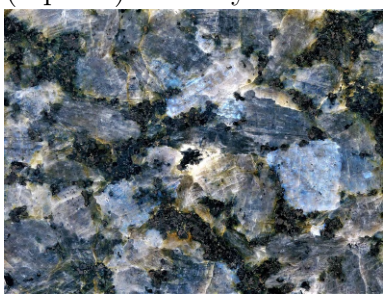
6. (1 point) Identify the following.



7. (1 point) Identify the following.

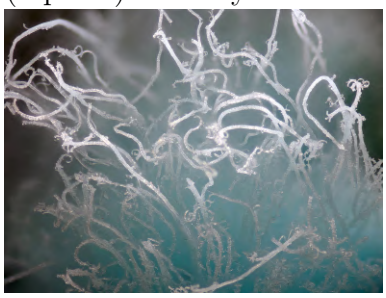


8. (1 point) Identify the following.



Station 11

1. (1 point) Identify the following.



2. (1 point) Identify the following.



3. (1 point) Identify the following.



4. (1 point) Identify the following.



5. (1 point) Identify the following.



6. (1 point) Identify the following.



7. (1 point) Identify the following.



8. (1 point) Identify the following.



Station 12

1. (1 point) Identify the following.



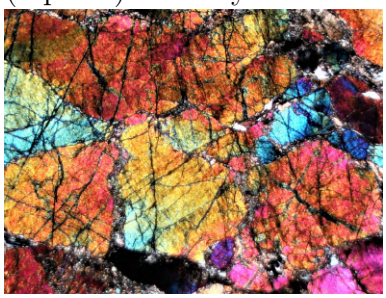
2. (1 point) Identify the following.



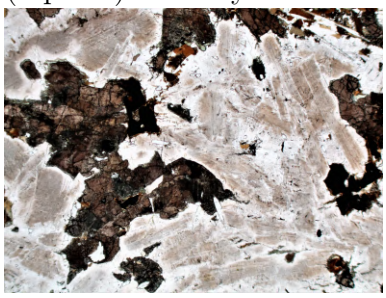
3. (1 point) Identify the following.



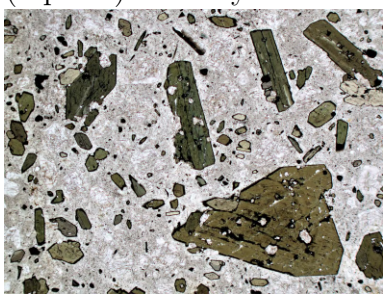
4. (1 point) Identify the following.



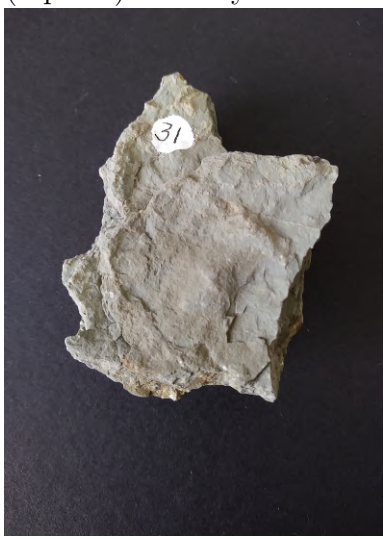
5. (1 point) Identify the following.



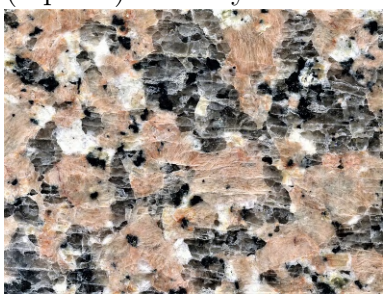
6. (1 point) Identify the following.



7. (1 point) Identify the following.



8. (1 point) Identify the following.



Station 13

1. (1 point) Identify the following.



2. (1 point) Identify the following.



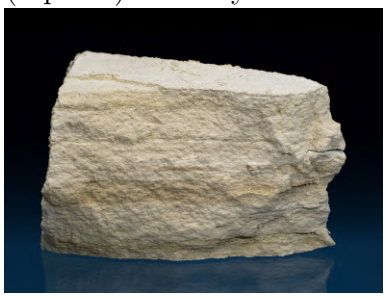
3. (1 point) Identify the following.



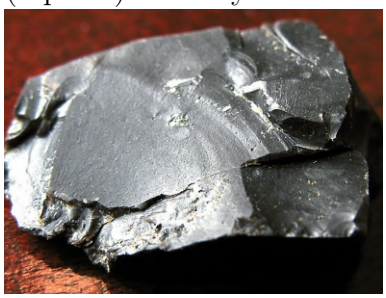
4. (1 point) Identify the following.



5. (1 point) Identify the following.

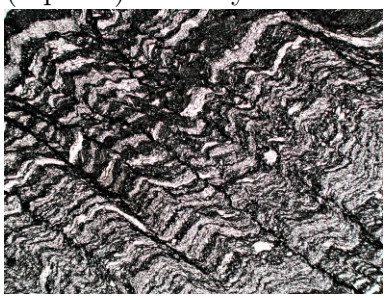


6. (1 point) Identify the following.



Station 14

1. (1 point) Identify the following.



2. (1 point) Identify the following.



3. (1 point) Identify the following.



4. (1 point) Identify the following.



5. (1 point) Identify the following.

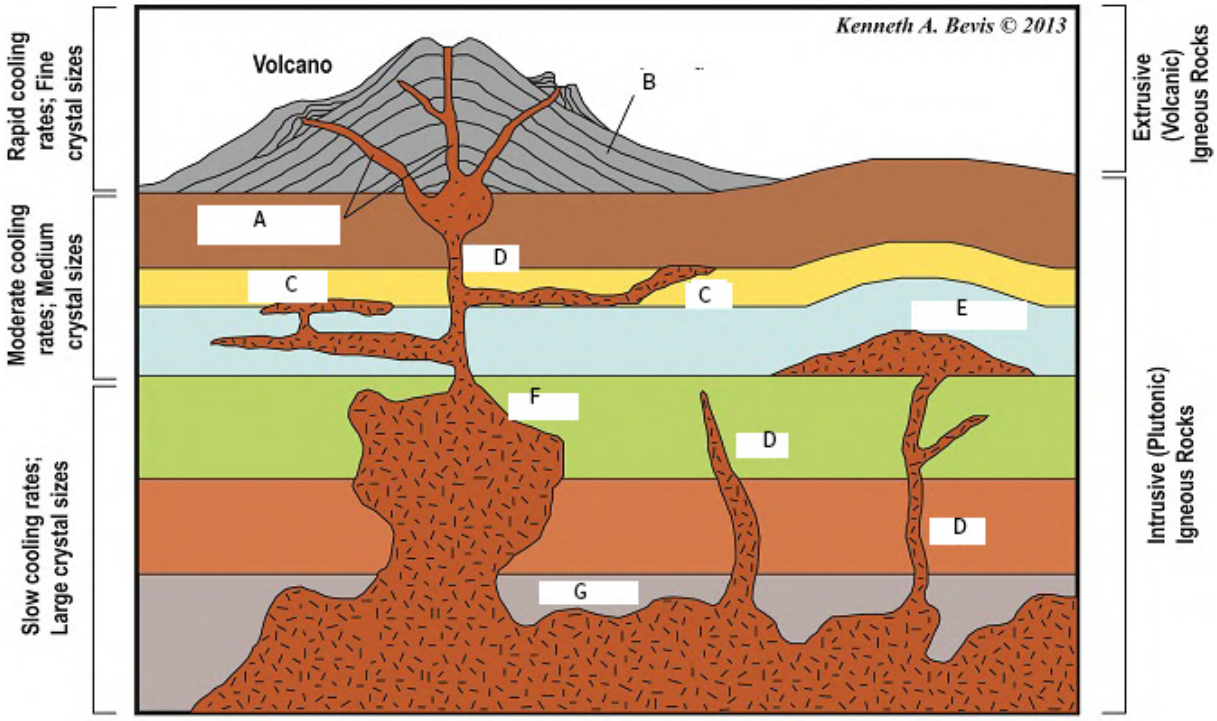


6. (1 point) Identify the following.



Station 15

1. (4 points) Label the following diagram.



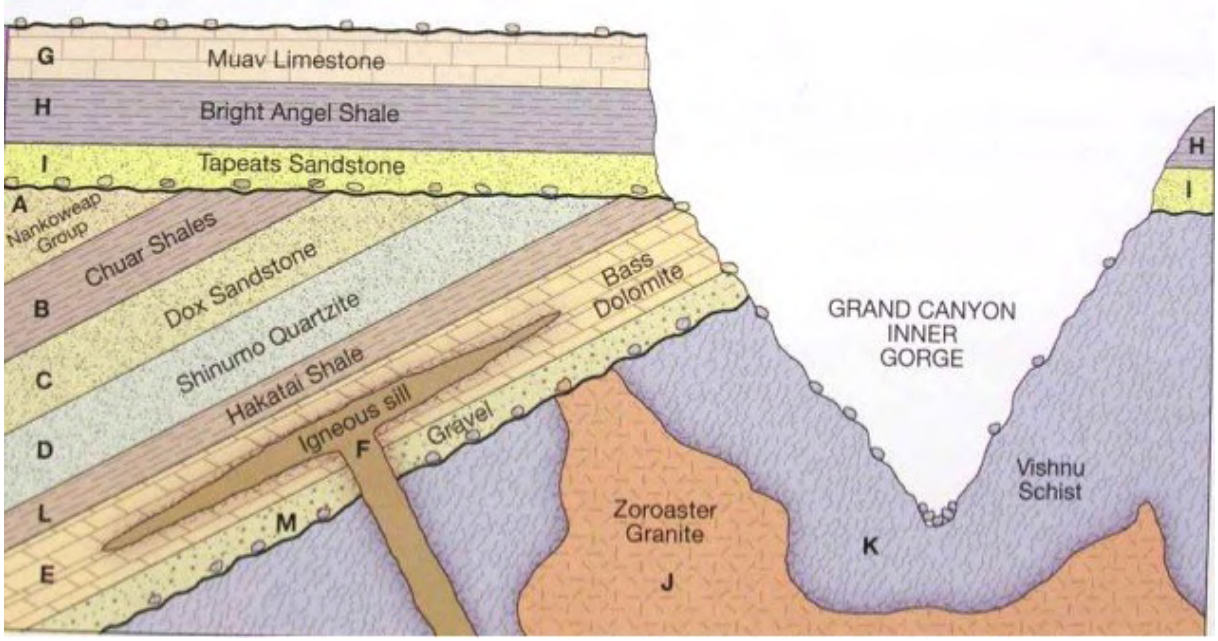
2. (1 point) Define what a xenolith is.

3. (1 point) You discover a tabular igneous body that is concordant with sedimentary bedding but has baked contacts above and below. What type of intrusion is it? **(Tiebreaker #7)**

Station 16

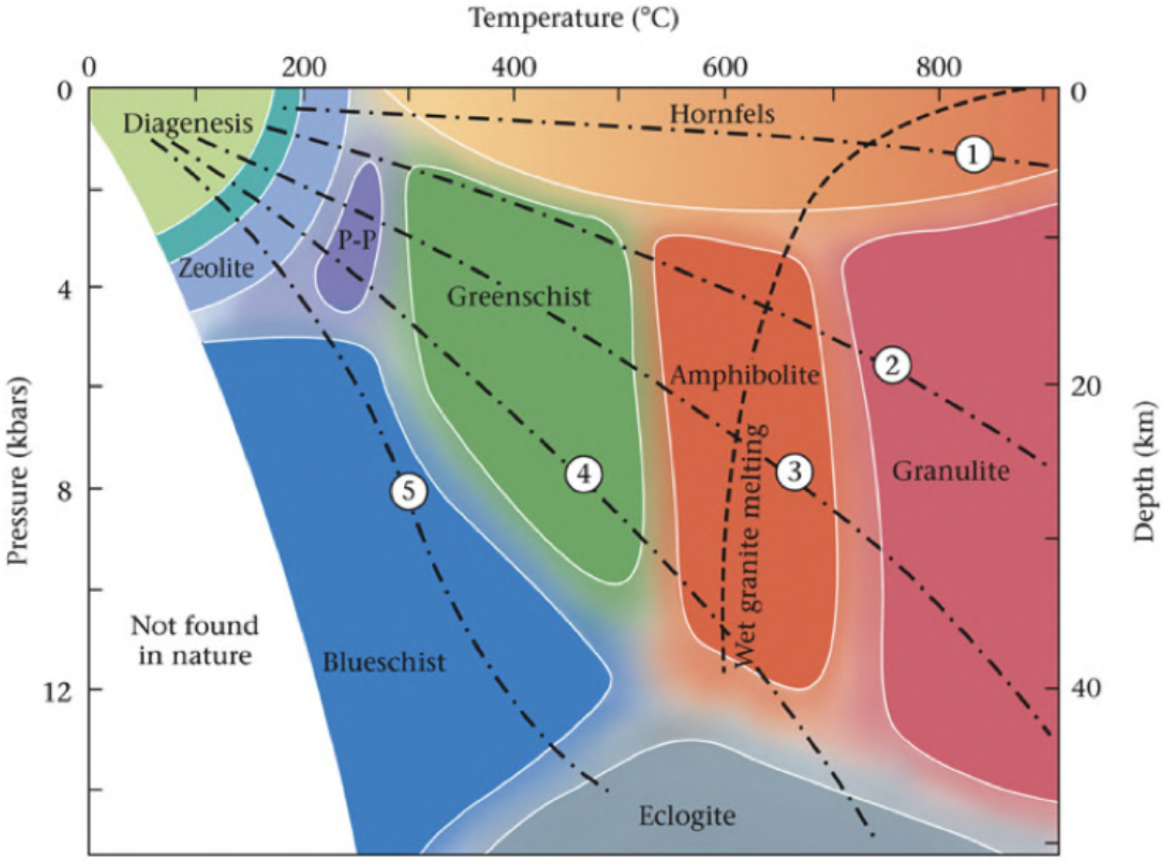
1. (2 points) Name and briefly explain the four major principles used in relative dating.

2. (8 points) Fill in the following diagram (Oldest to Youngest).



Station 17

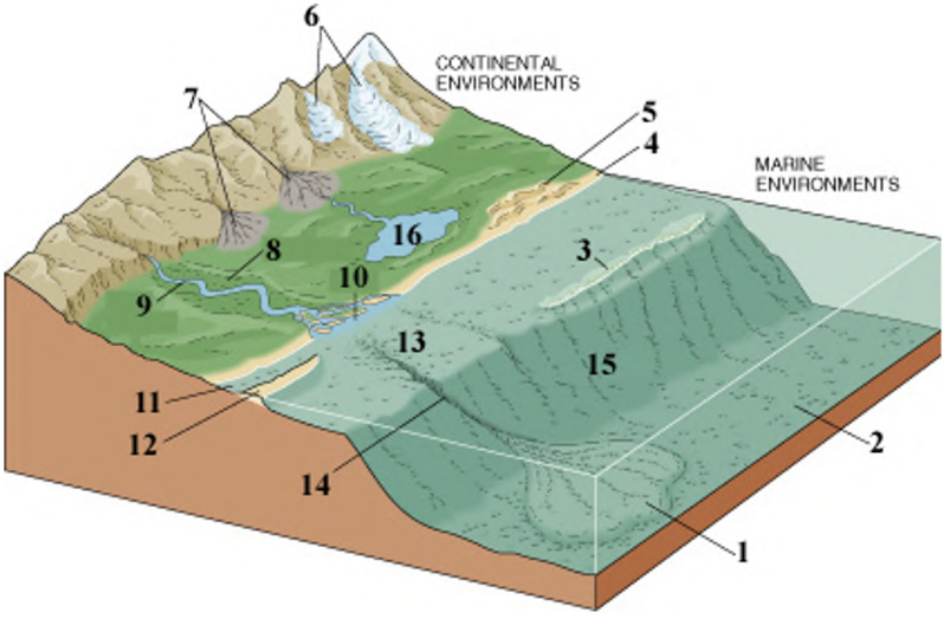
1. (5 points) Label the following diagram.



2. (3 points) Order the following metamorphic facies from lowest to highest grade.

Station 18

1. (8 points) Label the following.



Brown Invitational - Repaired practice edition

Division B | Rocks And Minerals | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

Name: _____ Date: _____

s2-10. (2 points) Authored replacement for source item s2-10: Supplied field notes place celestite with gypsum in evaporite beds. Describe an environment consistent with those observations.

Answer: _____
_____
_____

s2-11. (1 points) Authored replacement for source item s2-11: Supplied crystallographic data: chalcopyrite is tetragonal and celestite is orthorhombic. Do these two specimens belong to the same crystal system?

Answer: _____
_____
_____

s6-3. (2 points) Authored replacement for source item s6-3: Repair data: green amazonite is a variety of microcline; microcline is triclinic and orthoclase is monoclinic; both feldspars have composition KAlSi3O8. Explain which pair illustrates polymorphism.

Answer: _____
_____
_____

s6-5. (2 points) Authored replacement for source item s6-5: Can a green feldspar’s color or ultraviolet appearance alone establish that it contains lead? Describe a stronger test.

Answer: _____
_____
_____

s15-1. (4 points) Authored replacement for source item s15-1: Label these supplied descriptions: A is a sheet parallel to bedding; B cuts across bedding; C is a concordant dome-shaped intrusion; D is an intrusive body with exposed area greater than 100 square kilometres.

Answer: _____
_____
_____

s16-1. (2 points) Authored replacement for source item s16-1: Name and explain these four relative-dating principles: superposition, original horizontality, lateral continuity and cross-cutting relationships.

Answer: _____
_____
_____

s17-1. (3 points) Authored replacement for source item s17-1: Replace the missing facies diagram using this supplied schematic: A has moderate pressure and lower temperature; B has similar pressure and higher temperature; C has high pressure and low temperature. The supplied candidate facies are greenschist, amphibolite and blueschist. Match each once.

Answer: _____
_____
_____

s17-2. (3 points) Authored replacement for source item s17-2: Explain why hornfels, greenschist, blueschist, amphibolite and granulite cannot all be put into one universal lowest-to-highest temperature grade sequence without pressure information.

Answer: _____
_____
_____